HEMAL PATEL

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PROFESSIONAL SUMMARY

Senior Software Engineer with over 11 years of experience building multiplayer gameplay systems and designer- and artist-facing tools in C++ on Unreal Engine 5 (and Unity). Designs server-authoritative systems on top of Unreal GameMode and GameState data, from minigame scoring to analytics, and partners with producers, designers, technical artists and engine teams to give content creators the building blocks they need. Shipped live-service products on PC and the hardware-constrained Meta Quest 3.

TECHNICAL SKILLS

Engines: Unreal Engine 4/5, Unity (also Godot, Defold, Panda3D)

Languages: C++, C#, Python, Lua

Gameplay Engineering: Game Mode / Game State / Player State, Server-Authoritative Architecture, Multiplayer Networking & Replication, Gameplay Ability System (GAS), Gameplay Systems, UI, Multithreading, Performance Profiling & Optimization

Tools & Pipeline: Designer/Artist Editor Tools (Slate, UMG), CI/CD, BuildGraph, RenderDoc, Git, Perforce, Docker, Linux, JIRA

Platforms: PC (Windows, Linux), Meta Quest 3 (standalone VR)

Online & Backend: Meta XR SDKs, PlayFab, NATS, REST APIs, AWS, Grafana

AI Workflow: Agentic Tooling and Coding Workflows, Local LLM Deployment, Context Window Optimization

PROFESSIONAL EXPERIENCE

Gameplay Engineer (Freelance) **2026 – Present**

Ludomorphic *Remote (Horsham, PA)*

- Designed and built a multiplayer Railgun weapon in C++ and Blueprints, taking it from prototype to production system.
- Upgraded the Railgun to GMAS (GMC Ability System), an ability framework built on Unreal's General Movement Component, moving its behavior onto Gameplay Abilities.
- Built a Weapon Inventory system for weapon switching that incorporates GMAS Gameplay Abilities, structured to support the additional multiplayer weapon systems planned for the project.

Software Engineer II **Nov 2019 – Dec 2025**

WaveXR *Remote (Horsham, PA)*

- Helped architect and implement the Social Venue MiniGame manager, the platform's equivalent of a game mode: a server-authoritative system that holds and distributes points and drives a per-minigame leaderboard.
- Designed server-authoritative player-state systems and an analytics system built on Unreal GameMode data.
- Managed a team of four engineers and technical artists to deliver the Social Venue platform, including its minigame and live gameplay frameworks. Coordinated directly with producers and product stakeholders to define scope and technical roadmaps.
- Architected a custom Unreal-to-NATS messaging layer to handle real-time performer and attendee data serialization. Built accompanying UMG editor tools that allowed technical artists to configure data routing without writing code.
- Bypassed slow TeamCity CI/CD build loops by reverse-engineering Unreal's packaging pipeline to create a custom, UMG-based Editor Export Tool, letting content creators instantly generate local build artifacts.
- Designed a proxy system over NATS streams using a decoupled C++ MVVM pattern. This allowed runtime configuration overrides via Blueprints, isolating visual scripting from core systems and supporting cross-peripheral input switching.
- Profiled and eliminated critical RPC bottlenecks and Blueprint overhead using Unreal Insights. Achieved an 80% reduction in frame times and optimized runtime memory allocation.
- Rebuilt a high-draw Niagara particle system using the command pattern to optimize execution under full client load. Stabilized real-time performance on the hardware-constrained Quest 3, recovering frame rate from 15 fps to 36 fps with ASW enabled.
- Migrated legacy virtual environment codebases from Unity to UE5, porting core subsystems to native C++ to retain feature parity.

Unity Programmer (Contract) **Nov 2015 – Nov 2019**
Razor Edge Games *Remote (Horsham, PA)*

- Engineered core runtime mechanics in C#, including modular inventory frameworks and turn-based simulation systems designed for long-term scalability.
- Built an end-to-end Character Creator system handling complex mesh deformations, skeleton animation states, and UI sub-frameworks, reducing asset implementation time for design teams.
- Developed a custom beat-detection framework and an intuitive, script-configurable audio editor tool from scratch, so designers could synchronize runtime gameplay mechanics directly to audio tracks.

Unity Developer **Jun 2014 – Mar 2015**
Smartcloud Infotech Pvt. Ltd. *Pune, India*

- Developed and shipped three real-time interactive applications across different genres, adhering strictly to engine performance targets and project deadlines.
- Provided technical estimates and structured workflow recommendations during early discovery phases to establish stable engine pipelines.

ADDITIONAL EXPERIENCE

AI Trainer – Game Development Expert (Contract) **April 2026 – Present**
Handshake AI Fellowship *Remote (Horsham, PA)*

- Design and evaluate specialized technical prompts to test large language model capability in gameplay programming, systems design, and engine architecture.

Senior Staff Writer (Volunteer) **November 2016 – Present**
Gaming Cypher *Remote (Horsham, PA)*

- Write previews and reviews of AAA and indie titles, published at gamingcypher.com
- Attend conventions including PAX East for on-the-ground coverage of upcoming releases

TECHNICAL PROJECTS

ElementalArsenal: GAS Extension Plugin **C++, Unreal Engine 5, GAS, Editor Tools**

- Designed and implemented a modular plugin extending GAS to support custom equipment pipelines and automated elemental reactions, applying vector math and collision logic for spatial gameplay mechanics.
- Built custom Editor Tools to automate asset generation for GAS, accelerating the workflow for gameplay designers, using Slate and UMG.

NeoDataSync: Multiplayer Replication Plugin **C++, Unreal Engine, Multiplayer**

- Architected an Unreal Engine plugin to streamline replication of arbitrary data structures in multiplayer environments, utilizing clean C++ serialization wrappers to handle complex gameplay states.

NeoThreads: Unreal Multithreading Framework **C++, Unreal Engine, Multithreading**

- Developed a lightweight concurrency framework to safely offload heavy calculations from the Game Thread, improving frame stability and performance for CPU-intensive gameplay systems.

SHIPPED

Wave VR (Unity) | Wave Beta (Unreal) | Wave Standalone (Unreal) | Alimentary Avengers (Unity) | Tafe Planetary Plus (Unity) | Be Safe, Be Smart!
Notable Experiences: The Weeknd | Justin Bieber | Calvin Harris | John Legend | Tinashe | Pentakill

EDUCATION

DePaul University
Master of Science in Computer Game Programming (Coursework Completed)

CEPT University
Master of Business Administration, Information Technology & Project Management

Gujarat Technological University
Bachelor of Technology in Information Technology